Execution Speed: Programming Tips

While computer hardware and programming languages are important for maximizing execution speed, they are not something you change on a day-to day basis. In comparison, *how you program* can be changed at any time, and will drastically affect how long the program will require to execute. Here are three suggestions.

First, use integers instead of floating point variables whenever possible. Conventional microprocessors, such as used in personal computers, process integers 10 to 20 times faster than floating point numbers. On systems without a math coprocessor, the difference can be 200 to 1. An exception to this is integer division, which is often accomplished by converting the values into floating point. This makes the operation ghastly slow compared to other integer calculations. See Table 4-6 for details.

Second, avoid using functions such as: sin(x), log(x), y^x, etc. These transcendental functions are calculated as a series of additions, subtractions and multiplications. For example, the Maclaurin power series provides:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \cdots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \frac{x^{10}}{10!} + \cdots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots$$

EQUATION 4-3
Maclaurin power series expansion for three transcendental functions. This is how computers calculate functions of this type, and why they execute so slowly.

While these relations are infinite in length, the terms rapidly become small enough to be ignored. For example:

$$\sin(1) = 1 - 0.166666 + 0.008333 - 0.000198 + 0.000002 - \cdots$$

These functions require about ten times longer to calculate than a single addition or multiplication (see Table 4-6). Several tricks can be used to bypass these calculations, such as: $x^3 = x \cdot x \cdot x$; $\sin(x) \approx x$, when x is very small; $\sin(-x) = \sin(x)$, where you already know one of the values and need to find the other, etc. Most languages only provide a few transcendental functions, and expect you to derive the others by means of the relations in Table 4-7. Not surprisingly, these derived calculations are even slower.